

(4.1) Proveďte třírozměrnou regresi: zadanými body (x_i, y_i, z_i) v prostoru $\mathbb{R}^3$ proložte rovinu

$$z = ax + by + c$$

tak, aby byl součet čtverců $\sum_i (z_i - (ax_i + by_i + c))^2$ minimální. Uvažujte body

$(0, 0, 1), (1, 0, 1), (0, 1, 1), (1, 1, 2), (1, -1, 0), (-1, 1, 1)$.

$$ax + by + c = z$$

$$\begin{pmatrix} a & b & c & z \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 1 & -1 & 1 & 0 \\ -1 & 1 & 1 & 1 \end{pmatrix}$$

Metoda nejmenších čtverců

$$A^T A = \begin{pmatrix} 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 4 & -1 & 2 \\ -1 & 4 & 2 \\ 2 & 2 & 6 \end{pmatrix}$$

$$A^T W = \begin{pmatrix} 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 2 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$$

$$A^T A \hat{x} = A^T W$$

$$\begin{pmatrix} 4 & -1 & 2 & 2 \\ -1 & 4 & 2 & 4 \\ 2 & 2 & 6 & 6 \end{pmatrix} \sim \begin{pmatrix} 4 & -1 & 2 & 2 \\ -1 & 4 & 2 & 4 \\ 7 & 1 & 3 & 3 \end{pmatrix} \sim \begin{pmatrix} 4 & -1 & 2 & 2 \\ 0 & 15 & 10 & 18 \\ 0 & 5 & 10 & 10 \end{pmatrix} \sim \begin{pmatrix} 4 & -1 & 2 & 2 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & -20 & -12 \end{pmatrix} \sim$$

$$\begin{pmatrix} 4 & 0 & 0 & \frac{8}{5} \\ 0 & 1 & 0 & \frac{4}{5} \\ 0 & 0 & 1 & \frac{3}{5} \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & \frac{2}{5} \\ 0 & 1 & 0 & \frac{4}{5} \\ 0 & 0 & 1 & \frac{3}{5} \end{pmatrix}$$

$$a = \frac{2}{5}$$

$$b = \frac{4}{5}$$

$$c = \frac{3}{5}$$

$$\frac{2}{5}x + \frac{4}{5}y + \frac{3}{5} = z$$

Toto je hledaná rovina!